

Project MedSahayak

Empowering Rural Healthcare with Plain-Language Medical AI

Problem Statement ID	26113
Problem Statement Title	Accessibility and quality of public healthcare services, particularly in rural and underserved areas
Theme	MedTech / BioTech / HealthTech
PS Category	Software
Team Name	The Persimmons

PROPOSED SOLUTION

A health platform that transforms confusing medical reports into clear, actionable, and regional-language insights for every patient.

Key Capabilities

- **Plain-Language Simplification:** Converts complex medical reports, prescriptions, and X-rays into everyday conversational language.
- **Multilingual Voice Assistance:** Explains diagnosis and prescriptions in the patient's native regional language via audio.
- **Centralized Health Records:** Consolidates reports, X-rays, prescriptions, and doctor notes into one seamless portal.
- **Vitals Tracking:** Visualizes blood sugar, BP, and weight over time using simple intuitive graphs.
- **Smart Reminders:** Sends timely medicine & checkup alerts with missed-dose tracking.

What Makes It Different

- **Focus on Understanding:** Built around patient comprehension rather than simple document storage.
- **Always-Available AI Helper:** Provides instant answers to basic health queries without jargon.
- **Longitudinal Tracking:** Monitors patient health trends across multiple clinic visits over time.
- **Designed for Inclusion:** Specifically tailored for low-literacy, elderly, and rural users to eliminate travel & language barriers.

TECHNICAL APPROACH & ARCHITECTURE

Implementation Process

1. Capture

Patient scans or photos a medical report, prescription, or X-ray using any standard smartphone camera.

2. Simplify (OCR + NLP)

OCR extracts text, while NLP translates jargon into plain explanations of diagnosis, dosage, and precautions.

3. Screen (Image ML)

A CNN model screens X-rays for key anomalies and flags cases needing urgent specialist referral.

4. Communicate (Voice UI)

Summary is read aloud in regional languages using TTS, supporting low-literacy and elderly patients.

Technologies Used

PWA (HTML, CSS, JS) Python (Flask / FastAPI)
Tesseract OCR Google Cloud Vision Bhashini API
TensorFlow / PyTorch ResNet / EfficientNet
gTTS / Coqui TTS SQLite

Key Integrations

- **Bhashini:** Powers Indian regional language translation & TTS.
- **NIH Chest X-Ray Dataset:** Training data for image classification model.
- **Tesseract OCR:** Text extraction from printed reports.

FEASIBILITY & RISK MITIGATION

Feasibility Analysis

- **Technical:** Highly feasible utilizing mature OCR, LLM, and computer vision frameworks.
- **Operational:** Requires patient & doctor adoption via simplified onboarding.
- **Financial:** Low MVP setup cost with scalable cloud micro-services.
- **Legal:** Enforces medical disclaimer compliance ("not a direct diagnosis").
- **Market:** High demand across rural public healthcare centers.

Challenges & Mitigation Strategies

Challenge: Handwritten text & OCR errors

→ *Mitigation:* Focus on typed reports first; allow manual user text verification.

Challenge: Risk of misleading AI advice

→ *Mitigation:* Display original report alongside simplified text + clear disclaimers.

Challenge: Connectivity & low digital literacy

→ *Mitigation:* Offline-first PWA, voice-first UI, and large iconic navigation controls.

IMPACT & BENEFITS

Target Audience Impact

- **Demystified Healthcare:** Patients understand diagnoses and medication guidelines without confusion.
- **Adherence to Treatment:** Reminders decrease missed doses and promote timely follow-up visits.
- **Reduced Anxiety:** Instant audio explanations alleviate fear during post-clinic consultations.
- **Elderly Independence:** Voice UI enables independent management without relying on family members.

Broader Ecosystem Benefits

- **Economically:** Reduces unnecessary repeat consultations caused by report misunderstanding, saving travel costs.
- **Socially:** Bridges the digital and healthcare literacy gap across language communities.
- **Government & Policy:** Aligns with Digital Health Mission (ABDM) and public healthcare accessibility goals.

RESEARCH & REFERENCES

Key Technical & Governmental Portals

- **SIH 2026 Problem Statement Portal:** sih.gov.in (Problem Statement ID: 26113)
- **Bhashini Language AI Platform:** bhashini.gov.in (Government of India Multilingual AI initiative)
- **NIH Chest X-ray Dataset:** kaggle.com/datasets/nih-chest-xrays/data (Model training/validation)
- **Tesseract OCR Engine:** github.com/tesseract-ocr/tesseract

Quick Reference Note: Bhashini powers the multilingual translation and voice explanation features. The NIH Chest X-ray dataset serves as reference data for anomaly screening. Tesseract OCR reads printed reports, supporting the core text simplification pipeline for rural and underserved healthcare accessibility.